Technical Writing: Scientific Typesetting with LaTeX

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1 Introduction

LaTeX is a document preparation system widely used in academia and technical fields. It separates document formatting from content creation, allowing researchers and engineers to focus on clear writing.

2 Mathematical Typesetting

Technical documents frequently require precise mathematical expressions. For example, the quadratic formula is represented as:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{1}$$

Inline mathematical equations such as $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$ format cleanly alongside body text.

3 Key Advantages

- Deterministic compilation across all operating systems.
- Robust automated referencing and citation management via BibTeX.
- Professional typographical quality.